PROJECT LOGBOOK

Computer-Vision-based Approach to Detect Fatigue Driving

Course: CS-477 Computer Vision (Fall 2025)
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Project Duration: 8 Weeks (2 Phases)

- **Phase 1:** Literature Review & Proposal (Weeks 1-4)
- **Phase 2:** Implementation & Deployment (Weeks 5-8)

National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

PROJECT LOGBOOK CS-477 FALL 2025

Contents

1 PHASE 1: Literature Review & Proposal

Weeks 1–4

1.1 WEEK 1

Progress:

1. Team formed (Haida Asif, Aleesha Waqar, Ayesha Nahman)
2. Project topic finalized: Computer-Vision-based-approach-to-detect-fatigue-driving
3. Initial brainstorming on dataset availability, problem scope, NVIDIA Jetson Nano feasibility
4. Discussed driver fatigue detection methodologies: MTCNN, CNN-based classifiers, multi-modal fusion
5. Explored real-time processing requirements and embedded platform constraints

Meeting Minutes Detail:

1. Meeting Duration: 25 minutes
2. Agenda: Project topic selection and team role assignment
3. Discussion Points: Non-invasive fatigue detection approaches, Jetson Nano computational capabilities, real-time processing challenges, selection of fatigue indicators (EAR, PERCLOS, MAR)
4. Action Items: Collect research papers on fatigue detection, explore available datasets for eye and mouth state classification, review MTCNN architecture papers

Key Decisions:

1. Selected computer vision-based approach for driver fatigue detection using facial features
2. Decided to target NVIDIA Jetson Nano platform for embedded deployment
3. Agreed to use multi-modal fusion combining eye closure, yawn detection, and head pose estimation

1.2 WEEK 2

Progress:

1. Collected 7 research papers on driver fatigue detection systems and deep learning techniques
2. Identified key papers on MTCNN for face detection and CNN-based feature classification
3. Created GitHub repository with organized directory structure (DOC/, Data/, Results/, Src/)
4. Explored Eye Aspect Ratio (EAR), PERCLOS, and Mouth Aspect Ratio (MAR) metrics
5. Documented dataset sources: Kaggle Open/Closed Eyes Dataset and Yawn Dataset

Meeting Minutes Detail:

1. Meeting Duration: 20 minutes
2. Agenda: Literature review progress and system architecture planning
3. Discussion Points: Comparison of fatigue detection algorithms, pipeline architecture (face detection → landmark localization → feature extraction → classification → fusion), dataset preprocessing requirements
4. Action Items: Complete comprehensive literature review document, analyze state-of-the-art methods, document comparison of different fatigue metrics

Key Decisions:

1. Multi-index fusion strategy adopted combining EAR, PERCLOS, MAR, and head pose
2. Focus on real-time processing
3. Repository structure finalized with separate directories for documentation, data, models, and source code

1.3 WEEK 3

Progress:

1. Completed literature review analyzing MTCNN, CNN classifiers, and PnP head pose estimation
2. Analyzed state-of-the-art fatigue detection methods from IEEE papers
3. Documented Eye Aspect Ratio (EAR) formula and PERCLOS computation methodology
4. Reviewed Mouth Aspect Ratio (MAR) calculation for yawn detection
5. Explored rule-based multi-modal fusion techniques for combining multiple fatigue indicators
6. Created comparison table of different detection algorithms and their computational requirements

Meeting Minutes Detail:

1. Meeting Duration: 25 minutes
2. Agenda: Technical methodology selection and feasibility analysis
3. Discussion Points: MTCNN vs traditional face detection methods, CNN architecture selection for eye/mouth classifiers, temporal thresholding for blink and yawn duration, head pose estimation using facial landmarks
4. Action Items: Finalize system block diagram, document algorithm specifications, prepare project proposal draft

Key Decisions:

1. Selected MTCNN for face detection due to accuracy and speed balance
2. Decided to train separate CNN models for eye state and mouth state classification
3. Established fatigue detection criteria: prolonged eye closure, frequent yawning, abnormal head pose

1.4 WEEK 4

Progress:

1. Completed comprehensive project proposal with technical specifications
2. Documented system architecture: MTCNN → facial landmarks → EAR/MAR computation → CNN classifiers → multi-modal fusion
3. Created detailed timeline for implementation phase (Weeks 5–8)
4. Specified hardware requirements: Jetson Nano 4GB, CSI camera, 128GB storage, 5V 4A power supply
5. Documented software stack: Ubuntu 18.04, Python 3.6+, CUDA 10.2, cuDNN 8.0, TensorRT 7.1
6. Defined performance metrics: detection accuracy, inference time, false positive/negative rates

Meeting Minutes Detail:

1. Meeting Duration: 30 minutes
2. Agenda: Proposal finalization and supervisor review preparation
3. Discussion Points: System requirements documentation, performance benchmarks, deployment strategy, data preprocessing pipeline, model optimization techniques for embedded systems
4. Action Items: Submit proposal to Dr. Tauseef ur Rehman, prepare for Phase 2 implementation, setup development environment

Key Decisions:

1. Proposal submitted with comprehensive technical documentation
2. Established clear milestones for Phase 2: Week 5 (data preparation + training), Week 6 (model deployment), Week 7 (real-time testing), Week 8 (optimization + documentation)
3. Confirmed dataset sources and preprocessing requirements
4. Planned to use TensorRT for model optimization on Jetson Nano

2 PHASE 2: Implementation & Deployment

Weeks 5–8

2.1 WEEK 5

Progress:

1. Setup development environment: installed VS Code, Python 3.6, OpenCV, TensorFlow, Keras
2. Implemented data augmentation pipeline: rotation, flipping, brightness adjustment
3. Developed CNN architecture for eye state classifier

4. Developed CNN architecture for mouth state classifier: similar structure optimized for mouth features
5. Implemented MTCNN face detection pipeline using mtcnn library

Meeting Minutes Detail:

1. Meeting Duration: 25 minutes
2. Agenda: Model training progress and initial results review
3. Discussion Points: Data augmentation strategies, CNN architecture design choices, training hyperparameters (learning rate, batch size, epochs), validation accuracy vs training time trade-off
4. Action Items: Complete model training, save trained models (eye_model.h5, mouth_model.h5), implement facial landmark detection

Key Decisions:

1. Used depth-separable convolutions for computational efficiency on embedded platform
2. Applied early stopping and dropout to prevent overfitting
3. Saved best models based on validation accuracy for deployment
4. Confirmed MTCNN performs adequately for real-time face detection

2.2 WEEK 6

Progress:

1. Implemented facial landmark detection using dlib face predictor
2. Developed EAR (Eye Aspect Ratio) calculator using geometric formulas from facial landmarks
3. Developed MAR (Mouth Aspect Ratio) calculator for yawn detection
4. Implemented PERCLOS computation: percentage of time eyes closed
5. Integrated trained CNN models for eye and mouth state classification
6. Developed temporal thresholding for blink detection , yawn detection
7. Implemented head pose estimation using PnP (Perspective-n-Point) algorithm
8. Created complete fatigue detection pipeline combining all modules

Meeting Minutes Detail:

1. Meeting Duration: 20 minutes
2. Agenda: Feature extraction and integration progress
3. Discussion Points: EAR threshold selection , MAR threshold tuning, temporal frame counting strategies, head pose angle thresholds (pitch, yaw, roll)
4. Action Items: Test integrated pipeline to validate detection accuracy, prepare for Jetson Nano deployment

Key Decisions:

1. Confirmed EAR threshold,PERCLOS fatigue threshold and yawn detection criteria of 0.25 for eye closure detection based on empirical testing
2. Implemented rule-based fusion

2.3 WEEK 7

Progress:

1. Setup NVIDIA Jetson Nano: installed JetPack SDK 4.6, CUDA 10.2, cuDNN 8.0, TensorRT 7.1
2. Configured CSI camera module and tested video capture
3. Converted Keras models to TensorRT optimized format for faster inference
4. Preparing environment for complete model loading

Meeting Minutes Detail:

1. Meeting Duration: 30 minutes
2. Agenda: Jetson Nano deployment and real-time performance evaluation
3. Discussion Points: TensorRT optimization results, inference speed improvements, camera integration challenges, system latency measurements, detection accuracy on live feed
4. Action Items: Optimize frame preprocessing, fine-tune detection thresholds based on real-time testing, document performance metrics

Key Decisions:

1. TensorRT optimization reduced inference time
2. Confirm system meets real-time requirements
3. Adjusted lighting normalization preprocessing to improve robustness
4. Validated fatigue detection accuracy for eye closure and for yawn detection

2.4 WEEK 8

Progress:

1. Conducted comprehensive testing
2. Fine-tuned detection thresholds
3. Prepared final project documentation: README, code documentation, performance analysis report
4. Completed GitHub repository organization with all source code, models, and documentation

Meeting Minutes Detail:

1. Meeting Duration: 25 minutes
2. Agenda: Final testing, optimization, and documentation completion
3. Discussion Points: System robustness evaluation, false positive/negative analysis, practical deployment considerations
4. Action Items: Finalize documentation, prepare final report, organize code repository for submission

Key Decisions:

1. Confirm overall system accuracy
2. Documented limitations

END OF LOGBOOK